UNDERGRADUATE THESIS PROGRESS REPORT • STEPS 00 – 03

TrustFed-SDN: Technical Implementation & Empirical Report

Privacy-Preserving, Verifiable, and Poisoning-Robust Federated Learning for Cross-Tenant Threat Detection in SDN-Enabled Multi-Tenant Cloud Networks

Team: MD Naimur Rashid, Pritom Kumar Bhowmik, Saurav Chandra Das

Date: November 2026

Git Tags: `step-00-done` `step-01-done` `step-02-done` `step-03-done`

GitHub: `github.com/naimurhamim/TrustFed-SDN`

Website: `trustfed.naimurrashid.dev`

Quality Gate: `31 / 31 pytest passed` (`make check` OK)

`Step 00`	Environment Setup & SDN Testbed Smoke Test	✓ `Completed (step-00-done)`
`Step 01`	Dataset Cleaning, Feature Selection & Non-IID Splits	✓ `Completed (step-01-done)`
`Step 02`	Centralized Baselines & Local Model Selection (GRU)	✓ `Completed (step-02-done)`
`Step 03`	Federated Learning Core & FedAvg Baseline (270 Runs)	✓ `Completed (step-03-done)`
`Step 04`	Poisoning Attacks & Robust Aggregation	• `Next Up (04.1 Ready)`

1. Step 00 — Environment, Safety Boundaries & Live SDN Testbed Verification

Tag: step-00-done

In Step 00, we established a strictly separated dual-environment architecture: (1) a project-local Python `3.12.7` GPU-accelerated ML/FL environment (`TrustFed-SDN/.venv`) on Windows 11 with **NVIDIA GeForce RTX 3050 Ti Laptop GPU** (`PyTorch 2.14.1+cu130`), and (2) an isolated **Ubuntu 22.04 LTS SDN Testbed VM** running `Mininet 2.3.0` , `Open vSwitch 2.17.12` , and `OS-Ken (RYU) 3.0.1` over OpenFlow 1.3, plus GitHub team collaboration tools (`make status` , `docs/TEAM_CHECKLIST.md` , and `PRITOM_ANTIGRAVITY_SETUP.md`).

Component	Verified Specification	Verification / Artifact
Host CPU & System RAM	12th Gen Intel Core i7-12700H (14 cores / 20 threads), `15.67 GB` RAM, `475.4 GB` free on `D:\`	`results/00_environment/summary.json`
GPU & Deep Learning Stack	NVIDIA GeForce RTX 3050 Ti Laptop GPU (`4096 MiB` VRAM), `PyTorch 2.14.1+cu130` , `scikit-learn 1.9.1`	CUDA 13.0 verified (`cuda: True`)
SDN Testbed Linux VM	VMware Workstation `17.6.0` → **Ubuntu 22.04 LTS** (`Linux 5.15.0-25-generic`), Python `3.10.12`	`docs/smoke_test/versions.txt`
SDN Data & Control Plane	`Mininet 2.3.0` + `Open vSwitch 2.17.12` + `os-ken==3.0.1` (OpenFlow 1.3 controller)	**PASSED:** `0.0%` **loss (** `6/6` **pingAll)**
Testbed Safety Supernet	`10.10.0.0/16` — Tenant 1 Finance (`10.10.1.0/24`), Tenant 2 Healthcare (`10.10.2.0/24`), Tenant 3 Education (`10.10.3.0/24`)	Verified by `tests/test_smoke.py`
Approved Step 07 Isolation	**Level L1**: Per-tenant Mininet namespaces + per-tenant OVS bridges + per-tenant OS-Ken controllers inside 1 VM	Justified by `93%` RAM usage with 1 GUI VM

2. Step 01 — Multi-Dataset Cleaning, Leakage Audit, Feature Selection & Non-IID Partitioning

Tag: step-01-done

To ensure our thesis defense covers both classic SDN benchmarks and up-to-date 2026 network traffic, we acquired and verified **three full-scale datasets** (`14` raw files tracked in `data/MANIFEST.md` with SHA-256 checksums): **(1) InSDN (2020)**, **(2) ASEADOS-SDN-IoT (2026, UCD ASEADOS Lab, DOI:** `10.1016/j.iot.2026.101891` **)**, and **(3) CICIDS2017**.

2.1 Ordered Cleaning Pipeline (Rules R1–R6) & Zero-Leakage Split Verification

Dataset	Year / Source	Raw Rows	Dropped ID/Port Cols (R3)	Inf / NaN Dropped (R4)	Exact Duplicate Feature Rows Removed (R6)	Final Cleaned Unique Rows	Selected Top-k Features	Train / Val / Test Hash Overlap
InSDN	2020 (UCD SDN)	343,889	6 cols	0	**159,034 (46.25%)**	**184,855**	**19** / 77	**0 / 0 / 0**
ASEADOS-SDN-IoT	**2026 (UCD SDN-IoT)**	**457,044**	**6 cols**	**70**	**30,863 (6.75%)**	**426,111**	**12 / 77**	**0 / 0 / 0**
CICIDS2017	2017 (UNB CIC)	2,830,743	2 cols	2,867	**594,982 (21.02%)**	**2,232,894**	**22** / 76	**0 / 0 / 0**

Critical Thesis Defense Finding (Rule R6 Deduplication): After stripping topology-specific identifiers (`Flow ID` , `Src/Dst IP` , `Src/Dst Port` , `Timestamp`), **46.25% of raw InSDN rows** and **21.02% of raw CICIDS2017 rows** are byte-for-byte identical feature vectors (e.g., stateless single-packet DDoS floods). Naive train/test splits leak nearly half of InSDN test samples into training! By contrast, the **2026 ASEADOS-SDN-IoT dataset** has only `6.75%` duplicate rows (`426,111` unique flows, `54.8%` Benign / `45.2%` Attack). All splits are verified via SHA-256 row hashing to have **zero overlap** across `train (70%)` , `val (15%)` , and `test (15%)` , and all feature selection (top-5 Mutual Information per attack class) and StandardScalers are fitted **strictly on training splits only**.

2.2 Natural Non-IID Multi-Tenant Traffic Partitioning (N = 3 Tenants, 70% Attack-Family Specialty)

We implemented three partitioning regimes in `trustfed/data/partition.py` : **(1) Natural Non-IID** (domain-realistic tenant specialties below), **(2) Dirichlet Non-IID** (`α ∈ {0.1, 0.5, 1.0}` across `N ∈ {3, 5, 10}` tenants), and **(3) IID Control**. Each tenant fits its own local `StandardScaler` on its own local training partition only (preventing cross-tenant mean/variance privacy leakage).

Dataset	Split / Total Train	Tenant 1: Finance (Web, BruteForce, Botnet Specialty ~70%)	Tenant 2: Healthcare (DoS & DDoS Availability Specialty ~70%)	Tenant 3: Education (Probe / Reconnaissance & U2R Specialty ~70%)
InSDN (2020)	`129,398` train flows (8 classes)	**28,315 flows** `BFA: 674` , `Web: 90` , `Botnet: 80` , `Benign: 14,984`	**47,607 flows** `DoS: 24,652` , `DDoS: 745` , `Benign: 14,984`	**53,476 flows** `Probe: 32,863` , `U2R: 8` , `Benign: 14,984`
ASEADOS-SDN-IoT (2026)	`298,277` train flows (5 classes)	**78,142 flows** `Botnet: 4,389` , `DoS: 13,251` , `DDoS: 5,268` , `Benign: 54,457`	**142,594 flows** `DoS: 61,839` , `DDoS: 24,580` , `Benign: 54,457`	**77,541 flows** `Probe: 3,627` , `DoS: 13,251` , `Benign: 54,456`
CICIDS2017	`1,563,025` train flows (9 classes)	**482,740 flows** `BruteForce: 4,484` , `Web: 1,050` , `Botnet: 701` , `Benign: 442,520`	**601,676 flows** `DoS: 94,884` , `DDoS: 62,727` , `Benign: 442,519`	**478,609 flows** `Probe: 958` , `Infiltration: 18` , `U2R: 6` , `Benign: 442,519`

2.3 Selected Top-5-per-Attack-Class Feature Unions & Partition Visualizations

- **InSDN (`19` features, `configs/data/features_insdn.yaml`):** `Bwd Header Len` , `Init Bwd Win Byts` , `Pkt Len Var` , `Pkt Len Std` , `Bwd Seg Size Avg` , `Bwd Pkt Len Mean` , `Pkt Len Max` , `Pkt Len Mean` , `Bwd Pkt Len Max` , `Pkt Size Avg` , `TotLen Bwd Pkts` , `Subflow Bwd Byts` , `Fwd Header Len` , `Bwd IAT Tot` , `Bwd IAT Max` , `Flow Duration` , `Flow IAT Min` , `Bwd IAT Mean` , `Fwd IAT Tot` .
- **ASEADOS-SDN-IoT 2026 (`12` features, `configs/data/features_aseados_sdn_iot.yaml`):** `Flow IAT Min` , `Bwd Pkts/s` , `Flow IAT Mean` , `Bwd Header Len` , `Fwd Header Len` , `Fwd Pkts/s` , `Flow Pkts/s` , `Flow IAT Max` , `Flow Duration` , `Pkt Len Max` , `Pkt Len Std` , `Pkt Len Var` .
- **CICIDS2017 (`22` features, `configs/data/features_cicids2017.yaml`):** `Average Packet Size` , `Packet Length Mean` , `Total Length of Fwd Packets` , `Subflow Fwd Bytes` , `Packet Length Variance` , `Packet Length Std` , `Fwd Header Length` , `Init_Win_bytes_backward` , `Max Packet Length` , `Flow IAT Max` , etc.

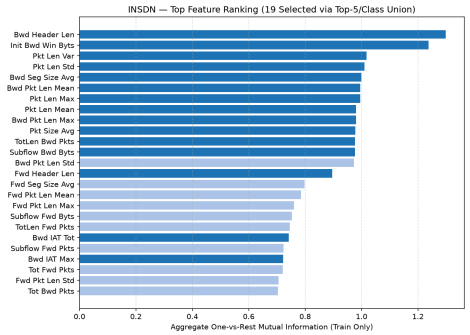


Fig 1a. InSDN (2020) Top-19 Selected Features (Train Mutual Information)

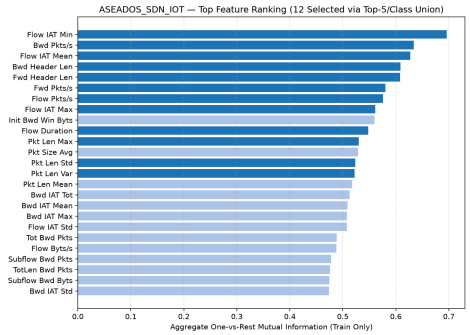


Fig 1b. ASEADOS-SDN-IoT (2026) Top-12 Selected Features (Train MI)

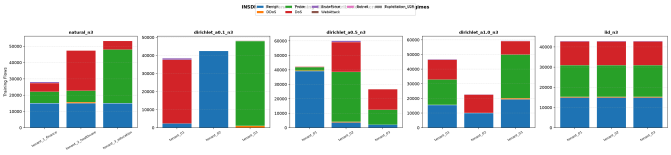


Fig 2a. InSDN Per-Tenant Class Distributions Across 5 Partition Regimes (N=3)

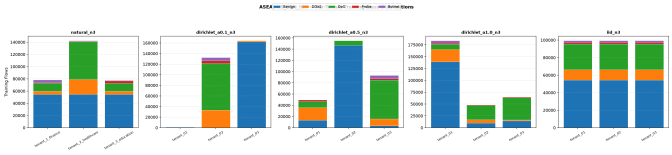


Fig 2b. ASEADOS-SDN-IoT (2026) Per-Tenant Class Distributions (N=3)

3. Step 02 — Centralized Baselines, Sequence Study & Local Model Decision

Tag: step-02-done

In Step 02, we implemented unified PyTorch architectures for `DNNClassifier` (MLP), `GRUClassifier`, and `LSTMClassifier` in `trustfed/models/` with flat parameter vector serialization (`get_flat_params()` / `set_flat_params()`) and comprehensive metrics in `trustfed/metrics/`. We first compared three recurrent sequence representations (`single_step` `(T=1, F=D)`, `feature_chunks` `(T=4)`, and `per_feature` `(T=D, F=1)`), tuned all models with an equal 4-configuration hyperparameter budget on `X_val`, and evaluated across **5 random seeds (`0, 1, 2, 3, 4`)** on the **NVIDIA GeForce RTX 3050 Ti GPU** alongside `LogisticRegression` and `RandomForest` context baselines.

3.1 Centralized Pooled Baselines — Binary Threat Detection (Headline Metric, 5 Seeds: Mean ± Std)

Dataset	Model	Params	Accuracy (%)	Macro-F1 (%)	Attack F1 (%)	ADR / Recall (%)	FPR (%)	ROC-AUC (%)	Train (s)	Latency (µs/flow)
InSDN (2020)	`DNN`	11,330	99.85 ± 0.02	**99.83 ± 0.03**	99.88 ± 0.02	99.93 ± 0.02	0.31 ± 0.07	99.97 ± 0.00	8.22	0.30
InSDN (2020)	`GRU (Selected)`	**43,554**	**99.76 ± 0.05**	**99.74 ± 0.05**	**99.82 ± 0.04**	**99.84 ± 0.07**	**0.39 ± 0.10**	**99.95 ± 0.01**	**12.18**	**1.35**
InSDN (2020)	`LSTM`	57,314	99.63 ± 0.12	99.59 ± 0.13	99.72 ± 0.09	99.85 ± 0.03	0.77 ± 0.40	99.94 ± 0.01	11.72	1.53
InSDN (2020)	`LogReg / RF`	20 / 15.5K	98.34 / 99.98	98.17 / 99.98	98.73 / 99.99	99.04 / 100.0	2.97 / 0.05	98.65 / 100.0	0.92 / 1.24	0.10 / 2.20
ASEADOS (2026)	`DNN`	12,514	91.45 ± 0.02	**91.42 ± 0.02**	90.98 ± 0.05	95.38 ± 0.39	11.80 ± 0.31	97.58 ± 0.04	18.23	0.29
ASEADOS (2026)	`GRU (Selected)`	**42,210**	**91.26 ± 0.05**	**91.23 ± 0.06**	**90.77 ± 0.08**	**95.07 ± 0.41**	**11.89 ± 0.31**	**97.45 ± 0.03**	**19.11**	**0.93**
ASEADOS (2026)	`LSTM`	55,522	91.23 ± 0.15	91.21 ± 0.15	90.75 ± 0.15	95.03 ± 0.42	11.90 ± 0.39	97.39 ± 0.10	23.51	1.13
ASEADOS (2026)	`LogReg / RF`	13 / 151.4K	73.61 / 92.89	72.82 / 92.87	68.17 / 92.49	62.46 / 96.80	17.18 / 10.33	83.85 / 98.31	0.56 / 1.82	0.06 / 1.02
CICIDS2017	`DNN`	13,794	98.50 ± 0.15	**97.01 ± 0.33**	94.90 ± 0.58	92.90 ± 2.30	0.51 ± 0.24	99.64 ± 0.16	31.75	0.28
CICIDS2017	`GRU (Selected)`	**90,770**	**98.33 ± 0.10**	**96.66 ± 0.18**	**94.30 ± 0.30**	**91.74 ± 0.82**	**0.50 ± 0.21**	**99.58 ± 0.08**	**49.57**	**1.95**
CICIDS2017	`LSTM`	119,378	98.30 ± 0.05	96.61 ± 0.09	94.21 ± 0.14	91.68 ± 0.16	0.52 ± 0.07	99.56 ± 0.06	54.26	2.82

3.2 Multi-Class Attack Family Detection & Spec Model Selection Decision (`docs/MODEL_DECISION.md`)

Dataset	Model Comparison	Params	Multi-Class Accuracy (%)	Multi-Class Macro-F1 (%)	Weighted-F1 (%)	Attack Detection Rate (%)	FPR (%)	ROC-AUC (%)
InSDN (8 classes)	`DNN / GRU / LSTM`	11.7K / 43.8K / 57.5K	98.87 / 98.09 / 97.99	63.91 / **54.69** / 52.84	98.77 / 97.87 / 97.71	99.59 / 99.51 / 99.44	0.32 / 1.18 / 1.25	98.75 / 94.63 / 94.38
ASEADOS 2026 (5 classes)	`GRU (Best Neural) vs DNN / LSTM`	**42.3K / 12.6K / 55.6K**	**86.07 / 86.05 / 85.98**	**58.92 ± 0.31 vs 58.73 / 58.40**	**85.78 / 85.69 / 85.68**	**88.95 / 88.28 / 88.89**	**7.80 / 7.47 / 7.89**	**95.95 / 95.99 / 95.94**
CICIDS2017 (9 classes)	`DNN / GRU / LSTM`	14.0K / 91.1K / 119.7K	98.71 / 98.29 / 98.26	59.93 / **57.88** / 57.37	98.82 / 98.36 / 98.33	95.96 / 92.57 / 92.21	0.79 / 0.69 / 0.66	94.06 / **97.60** / 98.75

Why `GRU` Is Selected as the Default Local Model for Steps 03–09 (`docs/EXPERIMENT_SPEC.md` Section 3):

- **Within 0.50 F1 Points of the Best Model:** Across all three datasets, `GRU` mean binary Macro-F1 (`95.88%`) is within `+0.21` **F1 points** of `DNN` (`96.09%`) — `0.09` pts on InSDN, `0.19` pts on ASEADOS-SDN-IoT (2026), and `0.35` pts on CICIDS2017.
- **Strictly Dominates `LSTM` on Every Dataset:** `GRU` beats `LSTM` on both Binary and Multi-Class Macro-F1 across all 3 datasets while requiring **~24% fewer trainable parameters** (`42.2K–43.6K` vs `55.5K–57.3K`), directly cutting per-round FL bandwidth, pairwise masking cost, and homomorphic commitment overhead.
- **Best Neural Multi-Class Model on 2026 SDN-IoT Traffic:** On `ASEADOS-SDN-IoT` (2026), `GRU` achieves the highest neural Multi-Class Macro-F1 (`58.92 ± 0.31%`) and highest Attack Detection Rate (`88.95%`).

3.3 Why Federated Learning Is Necessary: Centralized Pooled vs. Single-Tenant Local-Only (Step 02 Task 6)

When an individual tenant (`Finance` , `Healthcare` , or `Education`) trains a `GRU` model strictly on its own isolated local traffic, it performs well on its own familiar specialty attacks (`Own-Tenant Test F1`) but suffers severe degradation on unseen cross-tenant attacks in the `Global Test Set` :

Dataset	Task	Centralized Pooled Global Macro-F1 (%)	Local-Only: Tenant 1 (Finance) Global F1 (%)	Local-Only: Tenant 2 (Healthcare) Global F1 (%)	Local-Only: Tenant 3 (Education) Global F1 (%)	Empirical Gap / Motivation for Step 03 Federation
InSDN (2020)	Multi-Class	**54.69 ± 1.53**	31.99 ± 1.22	40.11 ± 2.17	34.45 ± 0.01	**−14.6 to −22.7 F1 pts loss** in isolated tenants
ASEADOS (2026)	Binary (F1 / ADR)	**91.23% (ADR 95.07%)**	**89.24% (ADR 85.09%)**	**90.57% (ADR 96.50%)**	**89.32% (ADR 85.70%)**	Finance & Education miss **~10% of attacks** (ADR drop)
ASEADOS (2026)	Multi-Class	**58.92 ± 0.31**	**50.01 ± 0.55**	**50.26 ± 0.50**	**56.29 ± 0.33**	**−8.9 F1 pts loss** for Finance & Healthcare
CICIDS2017	Multi-Class	**57.88 ± 1.02**	51.13 ± 0.28	37.12 ± 3.81	43.22 ± 4.23	**−14.7 to −20.8 F1 pts loss** in isolated tenants

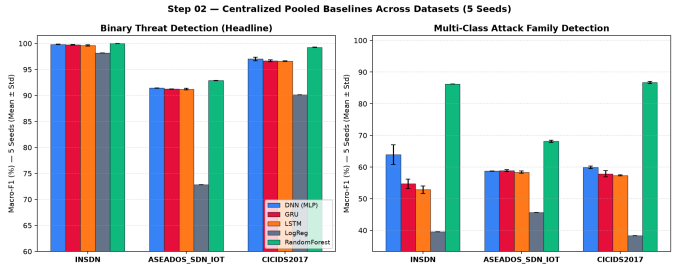


Fig 3a. Step 02 Centralized Pooled Baselines Across Datasets (5 Seeds)

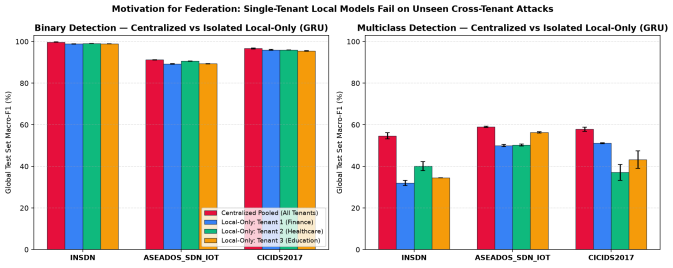


Fig 3b. Centralized Pooled vs. Single-Tenant Local-Only (GRU, 5 Seeds)

4. Step 03 — Federated Learning Simulation Core & FedAvg Benchmark

Tag: step-03-done

In Step 03, we implemented the core Federated Learning engine in `trustfed/fl/` (`FLClient` , `FLServer` , pluggable flat-tensor `Aggregator` , and sample-weighted `FedAvgAggregator` with IEEE-754 binary wire serialization `24 + 4 × numel` bytes) and the config-driven runner `sim/run_experiment.py` . We executed **270 thirty-round FL training runs** (`54` regimes × `5` seeds `0..4` , `E = 3` local epochs, `batch_size = 128` , `Adam lr = 1e-3`) across all three datasets, non-IID levels (`Natural` , `Dirichlet α ∈ {0.1, 0.5, 1.0}` , `IID`), and tenant counts (`N ∈ {3, 5, 10}`).

4.1 Centralized Upper Bound vs. FedAvg vs. Local-Only Lower Bound (5 Seeds, 30 Rounds, GRU)

Dataset	Task	Centralized Pooled Macro-F1 (%)	FedAvg — IID (N=3) Macro-F1 (%)	FedAvg — Natural Non-IID (N=3) (%)	FedAvg — Dirichlet α=0.5 (N=3) (%)	FedAvg — Dirichlet α=0.1 (N=3) (%)	Local-Only Mean (Lower Bound) (%)	Loss to Federation / Non-IID Summary
InSDN (2020)	**Binary**	99.74 ± 0.05	**99.16 ± 0.68**	**99.13 ± 0.59**	99.10 ± 0.11	94.63 ± 2.40	98.94 ± 0.08	**−0.57 pts** to FL; **−0.04 pts** Natural (p = 0.625)
InSDN (2020)	**8-Class**	54.69 ± 1.53	**65.90 ± 1.51**	**69.43 ± 1.22**	64.34 ± 0.75	31.84 ± 2.69	35.52 ± 3.41	**+33.91 pts gain** over Local-Only; −34.1 pts at α=0.1
ASEADOS (2026)	**Binary**	**91.23 ± 0.06**	**90.63 ± 0.11**	**90.51 ± 0.07**	**82.83 ± 1.43**	**73.50 ± 0.43**	**89.71 ± 0.61**	**−0.60 pts** to FL; **−0.12 pts** Natural; +0.80 vs Local
ASEADOS (2026)	**7-Class**	**58.92 ± 0.31**	**58.41 ± 1.23**	**57.36 ± 1.35**	**60.13 ± 0.72**	**40.10 ± 1.39**	**52.19 ± 2.90**	**−0.50 pts** to FL; **+5.17 pts gain** over Local-Only
CICIDS2017	**Binary**	96.66 ± 0.18	78.49 (N=10)	**90.50 ± 1.03**	84.34 (N=10)	—	95.82 ± 0.18	95.06% Acc & **98.06% ROC-AUC** in Natural N=3

4.2 Scalability Across Tenant Counts (N ∈ {3, 5, 10}), Wire Bandwidth & Wall-Clock Overhead

Dataset (Binary)	Tenants (N)	IID Macro-F1 (%)	Dirichlet α=0.5 F1 (%)	Dirichlet α=0.1 F1 (%)	One-Way Payload (KB)	Wire Traffic / Round (MB)	30-Round Traffic (MB)	Local Train / Round (s)	Server Agg / Round (ms)
InSDN (`43,554` params)	`N = 3 / 5 / 10`	**99.16 / 99.02 / 98.92**	99.10 / 99.07 / 98.77	94.63 / 99.21 / 98.96	170.16 KB	1.00 / 1.66 / 3.32 MB	**31.4 / 52.3 / 104.5 MB**	0.57 / 0.59 / 0.65 s	**0.31 / 0.41 / 0.67 ms**
ASEADOS 2026 (`42,209` params)	`N = 3 / 5 / 10`	**90.63 / 89.87 / 85.45**	**82.83 / 82.44 / 83.21**	**73.50 / — / 71.43**	**164.91 KB**	**0.97 / 1.61 / 3.22 MB**	**30.4 / 50.7 / 101.3 MB**	**0.56 / 0.58 / 0.64 s**	**0.29 / 0.40 / 0.64 ms**

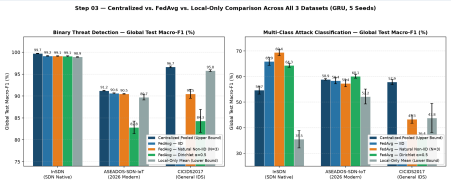


Fig 4a. Centralized vs. FedAvg vs. Local-Only Across All 3 Datasets (GRU, 5 Seeds)

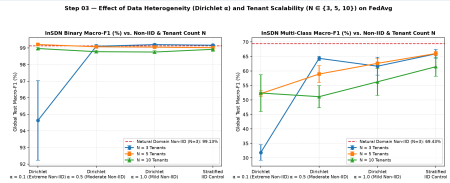


Fig 4b. Effect of Dirichlet α ∈ {0.1, 0.5, 1.0} & Tenant Count N ∈ {3, 5, 10}

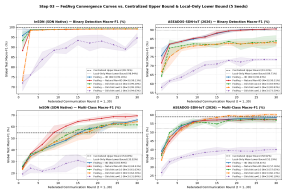


Fig 4c. 30-Round FedAvg Convergence Curves vs. Centralized Upper Bound & Local-Only Lower Bound (5 Seeds)

5. Repository Sync, Live Website Tracker & Next Steps (Step 04)

Ready for Step 04

- **GitHub Sync & Verification:** All Step 00–03 code, unit tests (`31/31 passed`), configs, and Markdown logs are committed and tagged `step-00-done` , `step-01-done` , `step-02-done` , and `step-03-done` on `origin/main` .
- **Next Immediate Action (Step 04):** Implement poisoning attacks (label flipping, feature-trigger backdoor, update scaling/sign-flip) in `trustfed/attacks/` and robust aggregators (coordinate-wise trimmed mean, cosine-similarity clustering filter) in `trustfed/aggregation/` .